

PHY 412: Statistical Physics

25-26 Semester II

Assignment – 2

Problems in black are for submission; Blue ones are for practice. Due date: Feb 24th before Exam

1. Consider N harmonic oscillators with coordinates and momenta $\{q_i, p_i\}$, and subject to a Hamiltonian $\mathcal{H}(\{q_i, p_i\}) = \sum_{i=1}^N \left[\frac{p_i^2}{2m} + \frac{m\omega^2 q_i^2}{2} \right]$.
 - (a) Calculate the entropy S , as a function of the total energy E .
(Hint. By appropriate change of scale, the surface of constant energy can be deformed into a sphere. You may then ignore the difference between the surface area and volume for $N \gg 1$. A more elegant method is to implement this deformation through a canonical transformation.)
 - (b) Calculate the energy E , and heat capacity C , as functions of temperature T , and N .
 - (c) Find the joint probability density $P(q, p)$ for a single oscillator. Hence calculate the mean kinetic energy and mean potential energy for each oscillator.
2. Consider a gas of N hard spheres in a box. A single sphere excludes a volume ω around it, while its center of mass can explore a volume V (if the box is otherwise empty). There are no other interactions between the spheres, except for the constraints of hard-core exclusion.
 - (a) Calculate the entropy S , as a function of the total energy E .
(Hint. $(V - a\omega)(V - (N - a)\omega) \approx (V - N\omega/2)^2$)
 - (b) Calculate the equation of state of this gas.
 - (c) Show that the isothermal compressibility, $\kappa_T = -V^{-1} \partial V / \partial P|_T$, is always positive.
3. N diatomic molecules are stuck on a metal surface of square symmetry. Each molecule can either lie flat on the surface, in which case it must be aligned to one of two directions, x and y , or it can stand up along the z direction. There is an energy cost of $\varepsilon > 0$ associated with a molecule standing up, and zero energy for molecules lying flat along x or y directions.
 - (a) How many microstates have the smallest value of energy? What is the largest microstate energy?
 - (b) For microcanonical macrostates of energy E , calculate the number of states $\Omega(E, N)$, and the entropy $S(E, N)$.
 - (c) Calculate the heat capacity $C(T)$ and sketch it.
 - (d) What is the probability that a specific molecule is standing up?
 - (e) What is the largest possible value of the internal energy at any positive temperature?
4. Consider the statistical mechanics of a tiny object with only two discrete states: one of energy E_1 and the other of higher energy $E_2 > E_1$.
 - (a) Find the ratio of the equilibrium probabilities p_1/p_2 to find our system in the two states, when weakly coupled to a heat bath of temperature T . What is the limiting probability as $T \rightarrow \infty$? And as $T \rightarrow 0$?
 - (b) Use the normalization of the probability distribution (the system must be in one or the other state) to find p_1 and p_2 separately. What is the average value of the energy E ?
5. Consider a polymer formed by connecting N elliptical molecules into a one-dimensional chain. Each molecule can align along either its long axis (of length $2a$) or short axis (length a). The energy of the monomer aligned along its shorter axis is higher by ε , that is, the total energy is $\mathcal{H} = \varepsilon U$, where U is the number of monomers standing up.
 - (a) Calculate the partition function, $Z(T, N)$, of the polymer.
 - (b) Find the relative probabilities for a monomer to be aligned along its short or long axis.
 - (c) Calculate the average length, $\langle L(T, N) \rangle$, of the polymer.
 - (d) Obtain the variance, $\langle L(T, N)^2 \rangle_c$.

6. A surface with N_0 adsorption centers has $N \leq N_0$ gas molecules adsorbed on it. Show that the chemical potential of the adsorbed molecules is given by $\mu = k_B T \ln(N/[(N_0 - N)a(T)])$, where $a(T)$ is the partition function of a single adsorbed molecule. Solve the problem by constructing the grand partition function as well as the partition function of the system.
[Hint: Neglect the intermolecular interaction among the adsorbed molecules.]
7. Consider an ideal gas of noninteracting particles in the grand canonical ensemble. The macrostate is given by $M \equiv (T, \mu, V)$ and the corresponding microstates $\{\vec{q}_1, \vec{p}_1, \vec{q}_2, \vec{p}_2, \dots\}$ have indefinite number of particles. Write down the grand partition function and grand potential. Then calculate P , N , and μ for this ideal gas.
8. Let us re-examine the generalized ideal gas introduced in the previous section, using statistical mechanics rather than kinetic theory. Consider a gas of N non-interacting atoms in a d -dimensional box of “volume” V , with a kinetic energy $\mathcal{H} = \sum_{i=1}^N A |\vec{p}_i|^s$, where $\vec{p}_i$ is the momentum of the i -th particle.
 - (a) Calculate the classical partition function $Z(N, T)$ at a temperature T . (You don’t have to keep track of numerical constants in the integration.)
 - (b) Calculate the pressure and the internal energy of this gas. (Note how the usual equipartition theorem is modified for non-quadratic degrees of freedom.)
 - (c) Now consider a diatomic gas of N molecules, each with energy $\mathcal{H}_i = A \left(|\vec{p}_i^{(1)}|^s + |\vec{p}_i^{(2)}|^s \right) + K \left| \vec{q}_i^{(1)} - \vec{q}_i^{(2)} \right|^t$, where the superscripts refer to the two particles in the molecule. (Note that this unrealistic potential allows the two atoms to occupy the same point.) Calculate the expectation value $\langle \left| \vec{q}_i^{(1)} - \vec{q}_i^{(2)} \right|^t \rangle$, at temperature T .
 - (d) Calculate the heat capacity ratio $\gamma = C_P/C_V$, for the above diatomic gas.
9. Consider rod-shaped molecules with moment of inertia I , and a dipole moment μ . The contribution of the rotational degrees of freedom to the Hamiltonian is given by $\mathcal{H}_{\text{rot}} = \frac{1}{2I} \left(p_\theta^2 + \frac{p_\phi^2}{\sin^2 \theta} \right) - \mu E \cos \theta$, where E is an external electric field. $\phi \in [0, 2\pi]$, $\theta \in [0, \pi]$ are the azimuthal and polar angles, and p_ϕ, p_θ are their conjugate momenta.
 - (a) Calculate the contribution of the rotational degrees of freedom of each dipole to the classical partition function.
 - (b) Obtain the mean polarization $P = \langle \mu \cos \theta \rangle$ of each dipole.
 - (c) Find the zero-field polarizability $\chi_T = \frac{\partial P}{\partial E} \big|_{E=0}$
 - (d) Calculate the rotational energy per particle (at finite E), and comment on its high and low-temperature limits.
 - (e) Sketch the rotational heat capacity per dipole.
10. Show that the Boltzmann entropy calculated using the condition of constant energy hypersurface and using a small width Δ about the hypersurface for a collection of noninteracting simple harmonic oscillators are same for all practical purposes in the thermodynamic limit.
11. Consider that N one-dimensional, independent, (quantum) harmonic oscillators with the Hamiltonian, $\mathcal{H} = \sum_{i=1}^N \hbar \omega \left(n_i + \frac{1}{2} \right)$ ($n_i \in \mathbb{W}$), are in thermodynamic equilibrium. (classical counting of states is valid in this quantum system)
 - (a) Find out the entropy.
 - (b) Hence, making use of the microcanonical ensemble, find the heat capacity in terms of temperature, and discuss its behavior for $k_B T \gg \hbar \omega$ and $k_B T \ll \hbar \omega$.
12. By letting two isolated systems exchange only volume (induced by, say, movable partition) to reach to an equilibrium for the combined system, find using the Laplace’s method of evaluating relevant integral

that in the thermodynamic equilibrium pressures in both parts are same. Repeat the problem for a hypothetical scenario where only number of particles are exchanged to show that at thermodynamic equilibrium, the chemical potentials are same as well.

13. Note that one can write the grand canonical partition function, Q , as $\sum_N z^N Z(N)$ where $z \equiv e^{\beta\mu}$ (called fugacity) and $Z(N)$ is the canonical partition function with fixed N particles.
 - (a) Verify that for ideal gas, $Z(N) = (Z_1)^N / N!$.
 - (b) Prove that for a grand canonical ensemble of ideal gas, the distribution of number of particles is Poisson distribution.
14. Consider a collection of N magnetic moments (each of magnetic moment μ_0) which orients either parallel or antiparallel to the applied external magnetic field B . The energy of a parallel configuration is $-\mu_0 B$ and that of the antiparallel one is $+\mu_0 B$. Making use of the “constant magnetic field ensemble”, find the Gibbs free energy and thence the (zero field) magnetic susceptibility, χ , when $k_B T \gg \mu_0 B$.
15. Consider an ideal gas atmosphere in a uniform gravitational force. Naturally, there is a temperature gradient but assume that each horizontal layer is in thermodynamic equilibrium.
 - (a) Find the chemical potential of the gas as a function of height z from the ground. Take acceleration due to gravity as $-g\hat{z}$.
 - (b) If there is no diffusion happening (*i.e.*, the gravity balances the diffusion tendency due to the upward density decrease), then using the result of part (a), find how $n(z)/n(0)$ (n being number density) varies with height. Assume temperature to be constant with height for the sake of analytical tractability of the calculation.
 - (c) Using the solution to part (b), find at which height the pressure of the gas reduces to $1/e$ -th of its value at the ground ($z = 0$)?